Department of Electrical and Electronics Engineering
Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: II Semester B.Tech AI&DS – 'A'
Course Code & Title: BE3251 - Basic Electrical and Electronics Engineering
Name of the Faculty member (s): S. Meenakshi Sundaravel

Innovative Practice Description

- **Unit / Topic:** Unit II / Construction and Working principle- DC Separately and Self excited Generators
- **Course Outcome:** CO 2
- **Unit Outcome:** TLO4
- **Activity Chosen:** Demonstration
- **Justification:**

The construction detail of the DC Generator is taught using the active learning approach of demonstration, Students' involvement in the learning is improved based on this demonstration approach.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.
At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about various parts of the DC machine for 2 minutes.
- ✓ Then I told them to Pair with their neighbors and discuss about the construction of the machine for another 1 minute.
- ✓ Finally, I shown the motor parts for each student and explained.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	2	1	-	-	-	-	-	-	-	1	-	1	-	1	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9
	1

Justification for correlation	The students can Function effectively as a team.

- **Images / Screenshot of the practice:**

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

Students told that it is good see all the parts of the machines individually and demonstration helps the students to understand the concepts easily.

❖ ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society
2. Students can able to explain the concepts in examination without any confusion.

❖ ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

1. Kothari DP and I.J Nagrath, "Basic Electrical and Electronics Engineering", Second Edition, McGraw Hill Education, 2020
2. S.K. Bhattacharya "Basic Electrical and Electronics Engineering", Pearson Education, Second Edition, 2017.

Signature of Faculty Member **HOD**

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Innovative Practice Description

- **Unit / Topic:** Unit III / Name and Symbol of various semiconductor devices
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 7
- **Activity Chosen:** One minute paper
- **Justification:**

  Students have to familiarize with the name, abbreviation and symbols of various semiconductor devices. Since, one minute paper activity provides a conceptual bridge between successive class periods. Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

  Time Allotted for the Activity: 02 minutes

- **Details of the Implementation:**

  At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement in the particular class.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	2	-	-	-	-	-	1	1	1	-	1	-	-	-

(1 – Low 2 – Moderate 3 – High)

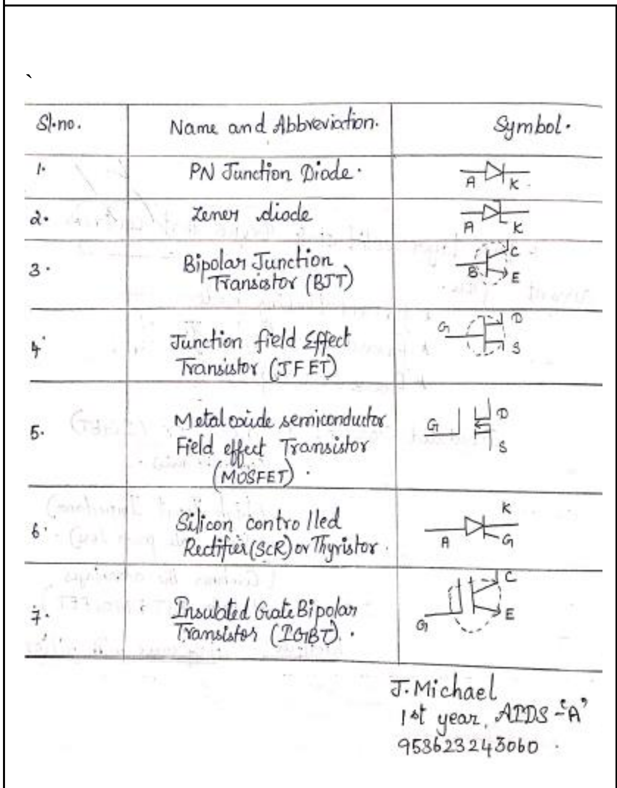
- **PO / PSO mapped:**

Innovative practice	**PO1** 3
Justification for correlation	The students can have the strong fundamental knowledge.

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution

Name and Symbol of various semiconductor devices – One minute paper



Sl.no.	Name and Abbreviation.	Symbol.
1.	PN Junction Diode.	A K
2.	Zener diode	A K
3.	Bipolar Junction Transistor (BJT)	C B E
4.	Junction field Effect Transistor (JFET)	G D S
5.	Metal oxide semiconductor Field effect Transistor (MOSFET).	G D S
6.	Silicon controlled Rectifier (SCR) or Thyristor.	K A G
7.	Insulated Gate Bipolar Transistor (IGBT).	C G E

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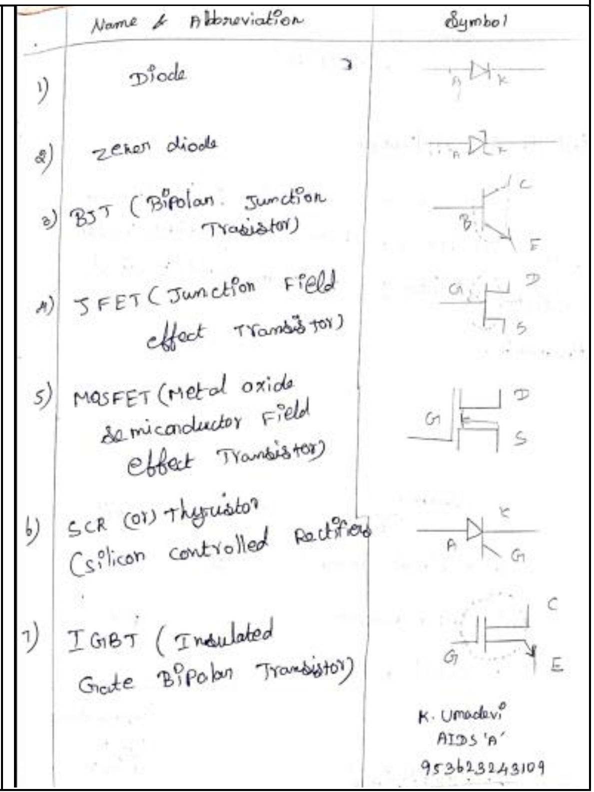

	Name & Abbreviation	Symbol
1)	Diode	A K
2)	zener diode	A K
3)	BJT (Bipolar Junction Transistor)	C B E
4)	JFET (Junction Field effect Transistor)	G D S
5)	MOSFET (Metal oxide semiconductor Field effect Transistor)	G D S
6)	SCR (or) Thyristor (silicon controlled Rectifier)	K A G
7)	IGBT (Insulated Gate Bipolar Transistor)	C G E

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- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
  - ❖ ***Benefit of the practice:***
    - ✓ Students can able to attend the question even in the questions are in indirect form.
    - ✓ Students can able to explain the concepts in examination without any confusion.
  - ❖ ***Challenges faced in implementation:***
    - ✓ Time utilization for conducting activity.

References:

1. Kothari DP and I.J Nagrath, "Basic Electrical and Electronics Engineering", Second Edition, McGraw Hill Education, 2020
2. S.K. Bhattacharya "Basic Electrical and Electronics Engineering", Pearson Education, Second Edition, 2017.

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Innovative Practice Description

- **Unit / Topic:** Unit III / Operation of Zener Diode
- **Course Outcome:** CO 3
- **Unit Outcome:** TLO7
- **Activity Chosen:** Virtual Lab
- **Justification:**

Virtual lab enables a virtual teaching and learning environment which develops students' practical knowledge. It is one of the most important eLearning tools that allow the student to perform various experiments without any constraints to place or time. Through this VLAB, the student can understand the operation, characteristics and application of zener diode as voltage regulator.

Time Allotted for the Activity: 15 minutes

- **Details of the Implementation:**

The students were asked to do the following steps

1. Set DC voltage to 10V
2. Set the Series Resistance (R_S) to 505 Ω
3. Set Zener voltage (V_Z) to 5V.
4. Vary the Load Resistance (R_L).
5. Voltmeter to be placed parallel to load resistor and ammeter in series with the series resistor.
6. Choose Load Resistance so that Zener diode is 'on' mode.
7. Note the Voltmeter and Ammeter readings for different values of Load Resistance.
8. Note the Load current(I_L), zener current(I_Z), Output voltage(V_O)
9. Calculate the voltage regulation.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	2	1	1	-	2	-	-	-	-	1	-	1	-	-	-

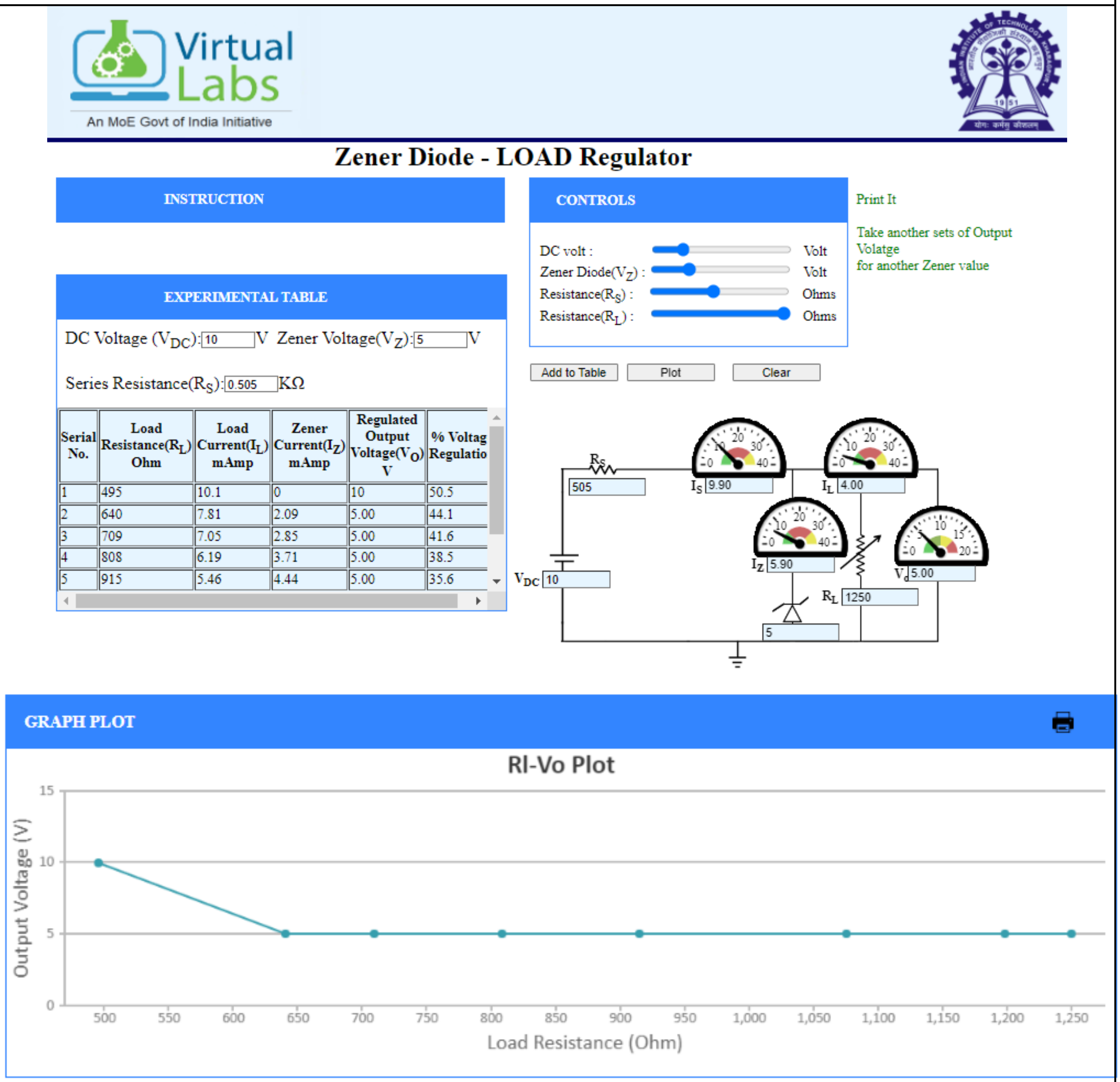
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO5**
	2
Justification for correlation	The students can effectively use the virtual lab and acquire knowledge about the characteristics.

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Operation of Zener Diode – Virtual Lab



- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***
  - ✓ More questions were asked to the students and came to know that they understood the concept well and the usage of virtual lab.
  - ✓ They could be able to plot V-I characteristics of zener diode.

- ***Benefit of the practice:***

  Beyond knowing the theoretical concepts, it is important to implement the concepts practically. So, knowing simulation is also needed to improve the technical knowledge of the students. Students can correlate the concepts studied theoretically with the experiments they simulated.

- ***Challenges faced in implementation:***

  I planned the activity for 15 minutes. But it takes 30 minutes to implement.

References:

1. David A. Bell , "Electronic devices and circuits", Oxford University higher education, 5th edition 2008.
2. S.K. Bhattacharya "Basic Electrical and Electronics Engineering", Pearson Education, Second Edition, 2017.

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Innovative Practice Description

- **Unit / Topic: Unit IV /** Conversion of number systems
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO11
- **Activity Chosen:** Flipped Classroom
- **Justification:**

A flipped classroom is a teaching method that shifts the traditional order of learning by having students learn content outside of class, and then apply it during class. This approach is designed to help students learn at their own pace, take responsibility for their learning, and engage in higher-level learning.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

  Flipped classroom activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. A student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.
  In this subject, Flipped classroom was conducted for the topic Conversion of number systems since some of the students have learned this topic earlier in their school education.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	2	1	-	-	-	-	-	-	-	1	-	1	-	1	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Conversion of number systems – Flipped Classroom


- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***

    Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

    The benefits of flipped classrooms are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information through their peers.

Challenges faced in implementation:

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and the peer leader will deliver the content to the group members. I planned the activity for 10 minutes only. But in real scenario it takes 15 minutes to complete this activity.

References:

1. Sedha R.S., "A textbook book of Applied Electronics", S. Chand & Co., 2008
2. James A .Svoboda, Richard C. Dorf, "Dorf's Introduction to Electric Circuits", Wiley, 2018.
3. A.K. Sawhney, Puneet Sawhney 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2015.

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